MatchY User Manual

Version 1.0 For Forensic Practitioners and Scientists

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1. Introduction

What is MatchY?

MatchY is a specialized forensic software tool designed to calculate **match probabilities** for Y-chromosome STR (Y-STR) profiles within pedigrees. It uses Monte Carlo simulation methods to assess the likelihood that a DNA profile belongs to a specific individual within a family tree.

Who Should Use MatchY?

- **Forensic Practitioners**: Case officers, forensic analysts, and investigators working with Y-STR evidence
- **Forensic Scientists**: Researchers and experts requiring statistical evaluation of Y-STR matchescases

Key Features

☑ **Two Operating Modes**:

- **Standard Mode**: Calculate match probabilities for a known suspect within a pedigree
- **Trace Mode**: Identify the most likely donor of an unknown Y-STR profile (trace) from pedigree members

☑ **Flexible Execution**:

- Browser-based GUI (simple, visual)
- Command-line interface (faster, multi-threaded)

☑ **Built-in Tools**:

- Pedigree Builder (visual family tree creator)
- Haplotype Editor (Y-STR profile manager)

☑ **Professional Reports**:

- PDF reports with visualizations
- Match probability tables
- Quality metrics and diagnostics

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2. Getting Started

System Requirements

- **Operating System**: Windows, macOS, or Linux
- **Python**: Version 3.8 or higher
- **RAM**: Minimum 4GB (8GB+ recommended for large pedigrees)
- **Storage**: 500MB free space

Installation

1. **Install Python** (if not already installed):
   - Download from python.org
   - Ensure "Add Python to PATH" is checked during installation
2. **Install MatchY dependencies**:

```
pip install -r requirements.txt
```

3. **Verify installation**:

```
streamlit --version
python main.py --help
```

Launching the GUI

```
streamlit run 1_🧬_Home.py
```

Your default web browser will open automatically. If not, navigate to `http://localhost:8501`

> **Note**: The GUI runs locally on your computer. No data is sent to external servers.

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3. Understanding the Science

3.1 What is Y-STR Analysis?

Y-chromosome Short Tandem Repeats (Y-STRs) are genetic markers inherited exclusively through the paternal line. Unlike autosomal DNA, Y-STRs:

- Pass from father to son virtually unchanged
- Undergo mutations at known rates (rapidly mutating (RM) Y-STRs have a mutation rate of >1% per generation)
- Allow tracing of paternal lineages over multiple generations

3.2 The Match Probability Problem

When a Y-STR profile from evidence matches (or partially matches) a reference profile, we need to answer:

> **"What is the probability that this profile belongs to a specific person within a family, given what we know about their relatives?"**

MatchY solves this by simulating thousands of possible scenarios through the family tree, accounting for:

- Mutation rates at each Y-STR marker
- The pedigree structure (who is related to whom)
- Known Y-STR profiles of family members
- The possibility of mutations occurring over generations

3.3 Weight of Evidence (Standard Mode)

What It Measures

The **weight of evidence** compares two hypotheses:

- **Hp (Prosecution Hypothesis)**: The suspect is the source of the evidence profile
- **Hd (Defense Hypothesis)**: Someone else (inside or outside the pedigree) is the source

Key Output Values

1. Inside Match Probability Distribution

This tells you: "*What is the probability that at least one other non-excluded, untyped individual within the pedigree shares the suspect's haplotype, given the pedigree structure and known haplotypes of typed relatives?*"

Interpretation: The higher the Inside Match Probability, the higher the likelihood that another individual matches the trace profile, and thus the lower the evidential weight against the suspect.

> 📌 **Forensic Note**: The match probability is highly dependent on the pedigree size and structure, as well as the number and position of typed individuals within the pedigree. Genotyping the father of the suspect (and indicating at least one allelic difference) is the best way to decrease the match probability and increase the evidential weight against the suspect.

2. Outside Match Probability

This represents: "*What is the probability that an unrelated male from (just) outside the current pedigree would match by coincidence?*"

> 📌 **Forensic Note**: In some cases it might be that the outside match probability is actually higher than the inside match probability. Especially in the trace mode this means that the true trace donor is more likely to be outside the current pedigree.

3.4 Trace Mode (Donor Identification)

What It Does

Trace mode answers a different question:

> **"Given an unknown Y-STR profile (trace), which person in this pedigree is the most likely source?"**

How It Works

1. The trace profile is compared against all possible positions in the pedigree
2. For each individual, MatchY calculates: "*What's the probability the trace came from this person?*"
3. Results are **normalized** so all probabilities sum to 100%

Key Output

Normalized Match Probabilities

Rank	Individual	Normalized Probability
1	John Smith	45.2%
2	Robert Smith	28.7%
3	Outside Pedigree	15.3%
4	Michael Smith	10.8%

Interpretation:

- **Rank 1 (John Smith, 45.2%)**: Most likely source, but not certain
- **Outside Pedigree (15.3%)**: ~1 in 7 chance the donor isn't in the pedigree at all
- **Low top probability**: Suggests markers can't discriminate well between relatives

3.5 Mutation Modeling

Single-Step vs Two-Step Mutations

Y-STR mutations typically involve:

- **Single-step**: Gain or loss of one repeat unit (most common)
- **Two-step**: Gain or loss of two repeat units (less common)

Two-Step Mutation Factor (default: 0.03)

- Controls the proportion of mutations that are two-step
- Based on published Y-STR mutation studies

- Affects calculations when evidence shows 2+ mutations from reference

> 📌 **Technical Note**: Unless you have population-specific mutation data, use the default value (0.03). This is based on large-scale studies of Y-STR mutation patterns.

Mutation Rates by Marker

Each Y-STR marker has its own mutation rate. MatchY uses:

- **Built-in databases**: Pre-loaded marker sets with published rates
- **Custom rates**: Upload your own mutation rate files if needed

Common marker sets:

- **Yfiler™**: 17 markers
- **Yfiler™ Plus**: 27 markers
- **PowerPlex® Y23**: 23 markers
- **RMplex**: 30 markers

> 📌 **Forensic Note**: The use of a kit with rapidly mutating (RM) Y-STR markers, such as RMplex, is adviced. Since the match probability is highly dependent on observed allelic differences within the pedigree, a marker set with elevated mutation rates increase the probability of observing such allelic differences. This does not, however, mean that the match probability will be higher when using a marker set with higher mutation rates - on the contrary actually: an allelic difference in a slowly mutating marker will result in a lower match probability (i.e. higher evidential value). The likelihood of observing a mutation in such a marker, however, is of course lower.

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4. Using the GUI (Graphical Interface)

4.1 Overview

The GUI is designed for users who prefer a visual, point-and-click interface. It's ideal for:

- Single-case analysis
- Learning how MatchY works
- Cases not requiring high-performance computing

4.2 Main Workflow

```
Upload Files → Set Parameters → Run Simulation → Review Results → Download Report
```

4.3 Step-by-Step Guide

Step 1: Select Marker Set

Location: Left sidebar

1. Click the **"Select marker set"** dropdown

2. Choose your Y-STR panel (e.g., "Yfiler", "PowerPlex_Y23", "RMplex")

> 📌 **What to Choose**: Select the marker set that matches your laboratory's Y-STR typing kit. If your kit isn't listed, you can create a custom marker set file (see Section 6.2).

Step 2: Upload Pedigree File

Location: Left sidebar

1. Click **"Upload pedigree file"**
2. Select a `.tgf` or `.ped` file

File Formats:

TGF (Trivial Graph Format) - Recommended for most users

```
1 GrandFather
2 Father
3 Son
4 Uncle
#
1 2
1 4
2 3
```

- Lines before `#`: Node definitions (ID, Name)
- Lines after `#`: Edges (Parent Child relationships)

PED (Pedigree Format) - Standard genetic format

```
Family1 1 0 0 1 0
Family1 2 1 0 1 0
Family1 3 2 0 1 0
```

- Columns: FamilyID, IndividualID, FatherID, MotherID, Sex, Phenotype
- Sex: 1=Male, 2=Female (MatchY only uses males)

> 💡 **Tip**: Don't have a pedigree file? Use the **Pedigree Builder** tool (see Section 6.1) to create one visually.

> ⚠️ **Important**:
>
> - Only **male** individuals are considered when opening .ped files
> - Use consistent names between pedigree and haplotype files
> - Ensure the pedigree forms a valid tree (no cycles, no disconnected branches)

Step 3: Upload Haplotypes File

Location: Left sidebar

1. Click **"Upload haplotypes file"**
2. Select a `.json` file containing Y-STR profiles

JSON Format:

```
{
  "GrandFather": {
    "DYS391": "10",
    "DYS389I": "13",
    "DYS439": "12",
    "DYS389II": "29"
  },
  "Father": {
    "DYS391": "10",
    "DYS389I": "13",
    "DYS439": "11",
    "DYS389II": "29"
  }
}
```

For Trace Mode, include a special "TRACE" key:

```
{
  "TRACE": {
    "DYS391": "10",
    "DYS389I": "13",
    "DYS439": "12",
    "DYS389II": "29"
  },
  "Father": {
    "DYS391": "10",
    ...
  }
}
```

Multi-Copy Markers (e.g., DYS385):

```
{
  "Individual1": {
    "DYS385": "11;14"
  }
}
```

- Use semicolon `;` to separate multiple copies
- Order doesn't matter (MatchY will sort them)

Intermediate Alleles (e.g., 13.2):

```
{
  "Individual1": {
    "DYS458": "17.2"
  }
}
```

> 💡 **Tip**: Use the **Haplotype Editor** (see Section 6.2) to create and validate your JSON files.

Step 4: Upload Files

Location: Left sidebar

1. After selecting both files, click **"Upload files"**
2. Wait for confirmation messages:
   - ☑ "Pedigree file uploaded successfully"
   - ☑ "Haplotypes file uploaded successfully"

The pedigree visualization will appear in the main area.

Step 5: Configure Simulation

Location: Main area

A. Select Mode

Standard Mode (Default):

- Uncheck "Use trace mode"
- Select the **suspect** from the dropdown (the individual whose profile matches the evidence)

> 📌 **Forensic Note**: Only a typed individual can be selected as suspect. This means that at least one typed individual should be present in the pedigree.

Trace Mode:

- Check "Use trace mode"
- Suspect selection becomes disabled (TRACE profile will be used instead)

> 📌 **When to Use Each Mode**:
>
> - **Standard Mode**: You have a suspect in mind and want to assess match probability
> - **Trace Mode**: You have unknown DNA and want to identify which pedigree member it came from

B. Exclude Individuals (Optional)

"Choose individuals to exclude from the simulation"

Use this when certain individuals are:

- Deceased before the crime
- Proven to be elsewhere (alibi)
- Eliminated by other evidence

> ⚠ **Important**: Excluded individuals are not considered in the match probability calculation, but their (simulated) haplotypes will still be used for the simulation process.

> 📌 **Forensic Note**: The match probability will usually decrease when excluding individuals (and therefor the weight of evidence against the suspect will increase). It is therefore adviced to use exclusion of individuals wisely.

C. Set Simulation Parameters

Click **"Set simulation parameters"** to expand advanced options:

Two-step mutation factor (Default: 0.03)

- Range: 0.00 - 1.00
- Controls proportion of two-step mutations
- **When to change**: Only if you have population-specific data; otherwise use default

Stability window (Default: 1000)

- Number of iterations where results must remain stable
- Higher = more confident results, longer runtime
- **Recommended**: 1000-5000 for most cases

Model validity (Default: 0.0100)

- Maximum allowed variation between convergence checks
- Lower = stricter convergence, more iterations
- **Recommended**: 0.01 for standard cases, 0.005 for complex pedigrees

Skip inside pedigree probabilities

- Check to disable inside pedigree calculations
- Only calculate outside match probability
- **Use when**: Only interested in outside match probability

Skip outside pedigree probabilities

- Check to disable outside match calculations
- Only calculate inside pedigree distribution
- **Use when**: Confident the source is within the pedigree

Use adaptive bias mode

- Automatically adjusts mutation bias based on convergence of the model
- **Recommended**: Use when analyzing very large pedigrees that converge slowly

> 📌 **For Most Cases**: Use the default values unless you have specific reasons to adjust them. The defaults are based on extensive testing of the software.

D. Choose Execution Mode

Execution Mode options:

1. Run in Browser (Single-threaded)

- Runs in your web browser
- Slower (uses one CPU core)
- Real-time progress updates
- **Best for**: Small pedigrees, learning, quick checks

2. Submit to CLI (Multi-threaded)

- Runs as separate process
- Much faster (uses multiple CPU cores)
- Shows output after completion
- **Best for**: Large pedigrees, production casework

If choosing CLI mode:

- Adjust **"Number of threads"** slider (default: 4)
- More threads = faster runtime (up to your CPU core count)

> 💡 **Performance Tip**: Runtime is dependent on a lot of factors: pedigree size and structure, number of typed and untyped individuals in the pedigree, number of typed markers and their mutation rates.

E. Name Your Simulation

"Give this simulation a name": Enter a case-relevant identifier

- Examples: "Case2024-0123", "Smith_Family_Analysis", "Trace_Donor_Case456"
- This name appears in the PDF report filename

"Your name": Enter the analyst's name

- Appears in the PDF report header
- Important for chain of custody and quality assurance

Step 6: Run Simulation

1. Click **"Start simulation"** (blue button)
2. Progress indicators will appear:
    - Browser mode: Real-time progress bars and log messages
    - CLI mode: Output log after completion

> ⚠ **Do Not**: Close your browser or navigate away during simulation. You'll see a warning message while it runs.

Step 7: Review Results

Standard Mode Results

Match Probabilities:

- Table showing the inside and outside match probability

Trace Mode Results

Most Likely Donor:

- Name and probability displayed prominently
- Color-coded severity indicator

Ranked Table:

- All individuals sorted by match probability
- Includes "Outside Pedigree" option
- Normalized to sum to 100%

Step 8: Download Report

1. Click **"Download simulation report"** (or "Download trace donor identification report")
2. PDF file saves to your downloads folder
3. Filename format: `[SimulationName]_report.pdf` or `[SimulationName]_trace_report.pdf`

Report Contents:

- Summary statistics
- Match probability tables/charts
- Known haplotypes table
- Pedigree diagrams
- Simulation parameters used
- Analyst name and date
- Quality metrics (convergence plots)

> 📌 **For Case Files**: This PDF is designed to be court-ready. It includes all necessary information for peer review and legal proceedings.

4.4 GUI Tips and Tricks

Tip 1: Use Example Files

- Click "Download example pedigree file in TGF format" in the sidebar
- Great for learning the file formats

Tip 2: Test with Simple Cases First

- Start with 3-5 individuals
- Add complexity once you understand the workflow

Tip 3: Check the Pedigree Visualization

- After upload, verify the tree structure looks correct
- Known individuals are highlighted

- Suspect has a distinct color

Tip 4: Save Your Configuration

- Keep your JSON haplotype files in a secure location
- Document your marker set choice
- Maintain version control for complex cases

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5. Using the CLI (Command Line)

5.1 Why Use the CLI?

The command-line interface is preferred for:

- **Batch processing**: Run multiple analyses automatically
- **High performance**: Full multi-threading support
- **Integration**: Embed MatchY in automated workflows
- **Reproducibility**: Script your exact analysis steps

5.2 Basic Usage

```
python main.py --config-path path/to/config.ini
```

5.3 Creating a Configuration File

Method 1: Generate from GUI

The easiest way to create a config file:

1. Use the GUI to set up your analysis
2. Choose "Submit to CLI (Multi-threaded)"
3. MatchY creates `streamlit_generated_config.ini` in the results folder
4. Use this file as a template for future CLI runs

Method 2: Manual Creation

Create a file named `config.ini`:

```
[pedigree]
path = data/pedigrees/smith_family.tgf
known_haplotypes = data/haplotypes/smith_profiles.json
marker_set = data/marker_sets/yfiler_plus.csv
suspect = Father
simulation_name = Smith_Case_2024
user_name = Dr. Jane Doe
exclude_individuals = Uncle,Cousin
two_step_mutation_factor = 0.10
stability_window = 1000
```

```
model_validity_threshold = 0.01
number_of_threads = 4
results_path = results/
bias = 0.1
```

Required Fields:

- `path`: Path to pedigree file (.tgf or .ped)
- `known_haplotypes`: Path to JSON haplotypes file
- `marker_set`: Path to CSV marker set file (or use GUI to generate)
- `simulation_name`: Name for this analysis
- `user_name`: Analyst name for report
- `results_path`: Directory to save results (created automatically)

Optional Fields:

- `suspect`: Individual name (omit for trace mode with TRACE in JSON)
- `exclude_individuals`: Comma-separated list of names to exclude
- `bias`: Mutation bias value (omit to disable)

Marker Set CSV Format:

```
marker,mutation_rate
DYS391,0.00251
DYS389I,0.00251
DYS439,0.00251
DYS389II,0.00251
DYS438,0.00074
```

5.4 Command-Line Options

Full syntax:

```
python main.py [OPTIONS]
```

Available Options:

Option	Short	Description	Default
`--config-path`	`-c`	Path to configuration file	`config.ini`
`--skip-inside`	`-i`	Skip inside pedigree calculations	Disabled
`--skip-outside`	`-o`	Skip outside pedigree calculations	Disabled
`--trace-mode`	`-t`	Enable trace mode	Disabled
`--adaptive-bias`	`-a`	Enable adaptive bias mode	Disabled

Examples:

Standard Analysis:

```
python main.py -c cases/case123.ini
```

Trace Mode:

```
python main.py -c cases/trace_analysis.ini --trace-mode
```

Skip Outside Probability (faster, inside-only):

```
python main.py -c cases/case123.ini --skip-outside
```

High-Performance Mode (adaptive bias, 8 threads):

```
python main.py -c cases/complex_pedigree.ini --adaptive-bias
```

(Set `number_of_threads = 8` in config.ini)

Skip Inside Probability:

```
python main.py -c cases/case123.ini --skip-inside
```

---

6. Helper Tools

6.1 Pedigree Builder

Location: Navigate to "📊 Pedigree Builder" in the sidebar

The Pedigree Builder is a visual tool for creating family trees without writing TGF or PED files manually.

Step-by-Step Guide

Step 1: Add Root Individual

1. Enter name in "Individual name" field
2. Click "Add individual"
3. This becomes the top of your pedigree (typically the oldest known ancestor)

Step 2: Add Children

1. Select parent from "Select parent" dropdown
2. Enter child's name
3. Click "Add child to selected parent"
4. Repeat for all children

Step 3: Build the Tree

- Continue adding generations downward
- The tool automatically creates parent-child relationships
- You can have multiple children per parent

Step 4: Visualize

- The pedigree diagram updates automatically
- Review the structure for accuracy

Step 5: Export

1. Click "Download pedigree as TGF"
2. Save the `.tgf` file
3. Use this file in the main MatchY interface

Example Walkthrough

Goal: Create a 3-generation pedigree

```
Grandfather (root)
├── Father
│   ├── Son1
│   └── Son2
└── Uncle
    └── Cousin
```

Actions:

1. Add "Grandfather" (root)
2. Select "Grandfather", add child "Father"
3. Select "Grandfather", add child "Uncle"
4. Select "Father", add child "Son1"
5. Select "Father", add child "Son2"
6. Select "Uncle", add child "Cousin"
7. Download TGF

> 📌 **Remember**: Only include males in the pedigree.

Common Issues

Issue: "Individual already exists"

- **Solution**: Each name must be unique. Use last names or suffixes (e.g., "John Sr.", "John Jr.")

Issue: Can't select a parent

- **Solution**: You must add at least one individual first. Start with the root.

6.2 Haplotype Editor

Location: Navigate to "🧬 Haplotype Editor" in the sidebar

The Haplotype Editor helps create and validate Y-STR profile JSON files.

Step-by-Step Guide

Step 1: Select Marker Set

- Choose the same marker set you'll use in the main analysis
- This loads the correct marker names

Step 2: Add Individuals

- Enter individual name (must match pedigree exactly)
- Click "Add individual"
- Repeat for all known profiles

Step 3: Enter Y-STR Data

For each individual:

1. Locate each marker row
2. Enter allele values in the input boxes

Single-Copy Markers (e.g., DYS391):

- Enter one value: `14`

Multi-Copy Markers (e.g., DYS385):

- Enter multiple values separated by semicolons: `11;14`
- Order doesn't matter

Intermediate Alleles (e.g., DYS458):

- Use decimal notation: `17.2`

Step 4: Add TRACE Profile (Optional)

For trace mode:

1. Click "Add TRACE profile"
2. A special "TRACE" individual appears
3. Enter alleles for the unknown profile

Step 5: Validate

- Click "Validate haplotypes"

- Checks for:
  - Missing markers
  - Invalid allele formats
  - Inconsistent multi-copy markers
  - Duplicate individual names

Step 6: Export

1. Click "Download haplotypes as JSON"
2. Save the `.json` file
3. Use in the main MatchY interface

Validation Rules

☑ **Valid Entries**:

```
14              (single copy)
11;14           (multi-copy, two copies)
11;13;15        (multi-copy, three copies)
17.2            (intermediate allele)
11.1;14.3       (multi-copy with intermediates)
```

❌ **Invalid Entries**:

```
14.5.2          (multiple decimal points)
abc             (non-numeric)
11, 14          (comma instead of semicolon)
11; 14          (space after semicolon)
```

Best Practices

1. Copy-Paste from Lab Reports

- Use spreadsheet software to organize data first
- Copy values column by column into the editor
- Always validate afterward

2. Use Consistent Naming

- Individual names in Haplotype Editor must match Pedigree Builder exactly
- Case-sensitive: "John" ≠ "john"

3. Handle Missing Markers

- MatchY doesn't allow missing values
- If a marker could not be analyzed in one individual, that marker should be removed from the marker set

4. Document Allele Calls

- Keep original lab reports alongside JSON files
- Note any ambiguous calls or stutter peaks

5. Trace Profile Quality

- Only include unambiguous loci in TRACE
- Exclude markers with mixtures or artifacts

---

7. Interpreting Results

7.1 Standard Mode Interpretation

Average pedigree probability

```
The average pedigree probability gives the user an indication on how likely the
current pedigree is that was provided. This value does not directly have an
influence on the weight of evidence. It can however give an indication on how
likely it is that the pedigree was correctly reconstructed. When the pedigree
contains impossible haplotype combinations (e.g. allelic differences that would
require three or more-step mutations) the average pedigree probability will be 0.
```

Extended average pedigree probability

```
Same as average pedigree probability, but for the extended pedigree (i.e. the
outside match probability pedigree).
```

Inside Match probability

Example: `0.0002` (or 0.02%)

What This Tells You:

This is the probability that at least one other non-excluded, untyped individual in the pedigree matches the suspect's haplotype.

Example Statement:

```
"The inside match probability of 0.0002 indicates that there is a probability of
approximately 0.02% that at least one other non-excluded, untyped individual in
the pedigree matches the suspect's haplotype."
```

Outside Match probability

Example: `0.0005` (or 0.05%)

What This Tells You:

This is the probability that a random individual (just) outside the provided pedigree matches the suspect's haplotype.

Example Statement:

```
"The outside match probability of 0.0005 indicates that there is a probability of
approximately 0.05% that a random individual (just) outside the provided pedigree
matches the suspect's haplotype."
```

> 📌 **Forensic Note**: Match probabilities depends on:
>
> - Pedigree structure
> - Number and location of typed individuals
> - Markers used (and their mutation rates)

The Convergence Plots

Your PDF report includes several convergence plots that show how the probabilities stabilize over iterations

- **Good convergence**: Line flattens out
- **Poor convergence**: Line continues oscillating

What to Look For: ☑ **Flat lines in the right half of the plot**: Good convergence, reliable results ❌ **Oscillating lines throughout**: Poor convergence, increase stability window ❌ **Sudden jumps**: When your pedigree contains large allelic differences, spikes might occur. Use adaptive bias

> 📌 **Quality Assurance**: Always review convergence plots before finalizing reports. Poor convergence means unreliable probabilities.

7.2 Trace Mode Interpretation

The Normalized Probability Table

Example output:

Rank	Individual	Original Probability	Normalized Probability
1	John Smith	4.52E-15	45.2%
2	Robert Smith	2.87E-15	28.7%
3	Outside Pedigree	1.53E-15	15.3%
4	Michael Smith	1.08E-15	10.8%

Understanding the Columns:

Original Probability:

- Raw probability from simulation
- Very small numbers (scientific notation)
- Not directly interpretable alone

Normalized Probability:

- Scaled so all probabilities sum to 100%
- **This is what you report**
- Represents relative likelihood among options

What This Tells You:

Scenario 1: High Inside Pedigree Probability

```
Interpretation: Strong indication that an individual inside the pedigree is the
most
likely source, though not conclusive proof.

Example Statement: "John Smith shows a normalized match probability
of 65.4%, indicating he is the most likely source of the trace DNA
among pedigree members. The next closest individual (Robert Smith, 22.1%)
is approximately 3 times less likely."
```

Scenario 2: High Outside Pedigree Probability

```
Interpretation: There's a substantial chance the true source is not
in the pedigree at all.

Example Statement: "While John Smith shows the highest match probability
among pedigree members (38.7%), the 'Outside Pedigree' option accounts
for 42.5% probability. This suggests the trace donor may be an unrelated
male or a paternal relative not included in the pedigree."
```

8. Best Practices

8.1 Quality Assurance

Pre-Analysis Checklist

Before running a simulation, verify:

- ☐ Pedigree visualizes correctly (no missing/extra individuals)
- ☐ All individual names match between pedigree and haplotype files
- ☐ Known individuals have complete Y-STR profiles (no missing markers)
- ☐ Marker set matches laboratory typing kit
- ☐ Suspect or TRACE is correctly identified

- ☐ Excluded individuals are appropriate for case context
- ☐ Simulation name is case-relevant and unique
- ☐ Analyst name is correct for report

Post-Analysis Checklist

After simulation completes, check:

- ☐ All convergence plots show stable lines
- ☐ No warning messages in console output
- ☐ Results make sense given pedigree structure
- ☐ PDF report contains all expected sections
- ☐ Known haplotypes table matches input data
- ☐ Simulation parameters in report match intent

Validation Run

For critical cases, consider:

1. **Run twice**:
    - The second run will have a different seed, thus resulting in slightly different results
    - Results should be nearly identical (within 1-2%)
    - Large differences indicate poor convergence
2. **Test with excluded individuals included**:
    - How much do results change?
    - Helps assess robustness

8.3 Performance Optimization

When to Adjust Parameters

Increase Stability Window (e.g., 1000 → 5000):

- Convergence plots show oscillation
- Results vary between runs
- Complex pedigree (many unknown individuals)

Decrease Model Validity (e.g., 0.01 → 0.005):

- Need higher precision for publication
- Close probabilities require better resolution
- Willing to wait for longer runtime

Enable Adaptive Bias:

- Standard analysis shows poor convergence
- Convergence plot shows lot of spikes (indicating large allelic differences)

Skip Inside or Outside:

- Testing/debugging (faster turnaround)
- Only need one type of probability
- Preliminary analysis before full run

8.4 Handling Difficult Cases

Problem: Poor Convergence

Symptoms:

- Convergence plots oscillate throughout
- Results vary significantly between runs

Solutions:

1. Increase stability window: 1000 → 5000 → 10000
2. Enable adaptive bias (if many known profiles)
3. Check for data entry errors in haplotypes
4. Simplify pedigree (remove distant relatives)

Problem: High Outside Pedigree Probability

Symptoms:

- Outside pedigree ranks #1

Solutions:

1. Verify pedigree completeness (missing relatives?)
2. Consider that the true donor may not be in the pedigree
3. Expand pedigree to include more distant relatives
4. Report honestly: "The donor may not be in this pedigree"

Problem: Very Long Runtime

Symptoms:

- Simulation runs for hours

Solutions:

1. Use CLI with more threads (10+)
2. Enable skip-outside or skip-inside for testing
3. Simplify pedigree (remove branches not relevant to case)
4. Consider running on more powerful hardware

---

9. Frequently Asked Questions

General Questions

Q: Is MatchY validated for casework?

A: The mathematical framework that MatchY is based on has been published in (ref). The implementation of the software tool "MatchY" has been published in (ref). The validation of the software tool has been published in (ref).

Q: How many individuals can MatchY handle?

A: Practical limits depend on your hardware. Generally speaking: around 10 unknown individuals should be doable on a home computer. More unknown individuals would require more computational power.

> 💡 **Tip**: For very large pedigrees, focus on the most relevant branches. Remove distant relatives unlikely to be involved in the case.

Q: Can I use MatchY for mixture interpretation?

A: No. MatchY assumes single-source Y-STR profiles. For Y-STR mixtures, deconvolution must be performed first using laboratory-validated mixture interpretation protocols. After deconvolution, if you obtain a clear major profile, that can be used as input to MatchY.

Q: Is MatchY open source?

A: MatchY's source code is available open source for transparancy. Feel free to contribute to the further development of MatchY.

Technical Questions

Q: What mutation model does MatchY use?

A: MatchY uses a **stepwise mutation model** with two-step mutation option:

- **Single-step**: +1 or -1 repeat (most common, ~97%)
- **Two-step**: +2 or -2 repeats (~3%, controlled by two-step mutation factor)

Q: Where do mutation rates come from?

A: Built-in mutation rates are compiled from peer-reviewed articles (refs).

Q: What's the difference between "inside" and "outside" probabilities?

A:

- **Inside**: Probability that at least one other non-excluded, untyped individual within the pedigree matches the suspect's haplotype.
- **Outside**: Probability that a random unknown individual (just) outside the provided pedigree matches the suspect's haplotype.

Q: Can I customize mutation rates?

A: Yes. Create a custom marker set CSV:

```
marker,mutation_rate
DYS391,0.00251
DYS389I,0.00251
YourCustomMarker,0.00300
```

Upload this via the GUI or reference it in `config.ini` for CLI.

> ⚠ **Caution**: Only change mutation rates if you have population-specific data. Using incorrect rates will produce misleading results.

Q: What does "adaptive bias" do?

A: Adaptive bias examines known individuals in the pedigree to detect consistent mutation directions:

- If multiple descendants show "up" mutations at a marker, that direction is favored
- If multiple descendants show "down" mutations, that direction is favored
- Improves accuracy when you have several known profiles spanning generations

When to use: When the pedigree contains large allelic differences between typed individuals

When NOT to use: If you want a more conservative (unbiased) analysis

---

Q: Why are my results slightly different each run?

A: MatchY uses Monte Carlo simulation, which is inherently random. Small differences (±1-2%) between runs are normal. Large differences indicate poor convergence. Increase the stability window and/or use adaptive biasing.

Q: How long should simulations take?

A: This is highly dependent on pedigree size and structure, number and position of typed and untyped individuals within the pedigree, number of typed markers and their mutation rates, the number of threads, etc...

Interpretation Questions

Q: What's a "good" match probability?

A: There's no universal threshold. A match probability of 50% corresponds to an Likelihood Ratio (LR) of 1: it is equally likely that the suspect is the only person within the pedigree carrying that haplotype, as it is that another person in the pedigree matches. Anything above 50% indicates that there is higher evidence that the suspect is the only carrier of that haplotype within the provided pedigree.

Always consider:

- Outside match probability
- Quality of input data
- Pedigree completeness
- Other case evidence

Q: Can Y-STR evidence prove someone is guilty?

A: **No.** Y-STR evidence, like all DNA evidence, provides statistical support for or against hypotheses. It doesn't prove guilt or innocence by itself.

Y-STR evidence should be presented as:

> "The Y-STR profile from the evidence is consistent with originating from [suspect] or a paternal relative. The statistical weight of this evidence is [describe probability]. This evidence should be considered alongside all other case facts."

Q: What if "Outside Pedigree" is the highest probability in trace mode?

A: This suggests the true donor is likely not in the pedigree. Report:

> "Analysis indicates the trace donor is more likely to be an unrelated male (or a paternal relative not included in the pedigree) rather than one of the known pedigree members. This suggests investigative leads should consider individuals outside this family."

Expand the pedigree to include more distant relatives

---

10. Troubleshooting

File Upload Issues

Problem: "Invalid pedigree file format"

Cause: File doesn't match TGF or PED format

Solution:

1. Open file in text editor
2. Check format matches examples in Section 4.3
3. Common issues:
    - Missing `#` separator in TGF files
    - Extra spaces or tabs
    - Windows vs Linux line endings (use `dos2unix` if needed)

Fix for TGF:

```
# Correct:
1 Father
2 Son
#
1 2

# Incorrect (no separator):
1 Father
2 Son
1 2
```

Problem: "Individual [Name] not found in pedigree"

Cause: Name mismatch between pedigree and haplotype files

Solution:

1. Compare names in both files (case-sensitive!)
2. Check for:
    - Extra spaces: "John " vs "John"
    - Different capitalization: "John" vs "john"
    - Special characters: "José" vs "Jose"
3. Edit files to make names exactly match

Problem: "Marker [Name] not found in marker set"

Cause: Haplotype file contains markers not in selected marker set

Solution:

1. Verify marker set matches your typing kit
2. Check marker name spelling (case-sensitive)
3. Common issues:
    - "DYS389-I" vs "DYS389I" (hyphen differences)
    - "DYSS391" vs "DYS391" (typo)

Problem: "Number of copies mismatch for marker [Name]"

Cause: Different individuals have different numbers of alleles for a multi-copy marker

Solution:

1. Check all entries for that marker in JSON file
2. Example problem:

```
"Individual1": {"DYS385": "11;14"},      // 2 copies
"Individual2": {"DYS385": "11;14;16"}    // 3 copies ❌
```

3. Verify laboratory data—this may indicate a data entry error

---

Problem: Simulation runs forever (hours)

Cause: Very slow convergence or infinite loop (rare)

Solution:

1. Cancel (Ctrl+C in CLI, or "Stop simulation" in GUI)
2. Reduce stability window: 5000 → 1000
3. Use CLI with more threads
4. Skip inside or outside for faster testing
5. Simplify pedigree (remove distant branches)

Problem: Results vary wildly between runs

Cause: Poor convergence or insufficient iterations

Solution:

1. Increase stability window significantly (e.g., 10000)
2. Decrease model validity (more strict)
3. Check convergence plots—if oscillating, need more iterations
4. Variation of ±1-2% is normal; ±10%+ indicates a problem

Problem: "Memory error" or "Out of memory"

Cause: Pedigree too large for available RAM

Solution:

1. Close other programs
2. Reduce number of threads (uses less memory per thread)
3. Simplify pedigree
4. Run on machine with more RAM
5. Use skip-inside or skip-outside (reduces memory usage)

Appendix A: Glossary

Allele: A specific value (repeat count) at a Y-STR marker (e.g., "14" at DYS391).

Haplotype: The full set of alleles across all Y-STR markers for an individual.

Intermediate Allele: An allele with fractional repeats (e.g., 17.2), typically due to incomplete repeat units.

Inside Match Probability: Probability that at least one other non-excluded, untyped individual within the provided pedigree matches the suspect's haplotype.

Mutation: Change in repeat count passed from father to son.

Outside Match Probability: Probability that a random individual (just) outside the provided pedigree matches the suspect's haplotype.

Pedigree: Family tree showing paternal relationships.

Stability Window: Number of simulation iterations required to have stable results before stopping.

Suspect: In standard mode, the individual whose profile is being evaluated.

TRACE: In trace mode, the unknown Y-STR profile being compared against pedigree members.

Two-Step Mutation: A mutation involving gain or loss of 2 repeat units (less common than single-step).

Y-STR: Y-chromosome Short Tandem Repeat marker; inherited paternally.

Appendix B: Quick Reference Cards

GUI Quick Start

1. **Select marker set** (sidebar)
2. **Upload pedigree** (.tgf or .ped file)
3. **Upload haplotypes** (.json file)
4. Click **"Upload files"**
5. **Configure simulation**:
    - Select suspect OR enable trace mode
    - Set execution mode (browser/CLI)
    - Name simulation
6. Click **"Start simulation"**
7. **Download report** when complete

CLI Quick Start

1. Create `config.ini` with required fields
2. Run: `python main.py -c config.ini`
3. Wait for completion
4. Find PDF report in `results/` folder

Trace Mode Checklist

- ☐ JSON file contains `"TRACE"` key with haplotype
- ☐ "Use trace mode" checkbox enabled (GUI)
- ☐ OR `--trace-mode` flag used (CLI)
- ☐ Report shows normalized probabilities table

- ☐ TRACE column appears in known haplotypes table

Troubleshooting Flowchart

```
Problem?
├─ Upload fails
│  ├─ Check file format (TGF/PED for pedigree, JSON for haplotypes)
│  └─ Verify names match between files
│
├─ Simulation won't start
│  ├─ Check suspect is selected (or trace mode enabled)
│  └─ Verify all individuals have haplotype data
│
├─ Simulation runs forever
│  ├─ Reduce stability window
│  └─ Use CLI with more threads
│
├─ Poor convergence
│  ├─ Increase stability window
│  └─ Enable adaptive bias
│
└─ Report issues
   ├─ Update WeasyPrint
   └─ Check results/ folder for plot images
```

Appendix C: Example Files

Example Pedigree (TGF)

File: `example_pedigree.tgf`

```
1 Grandfather
2 Father
3 Uncle
4 Son1
5 Son2
6 Cousin
#
1 2
1 3
2 4
2 5
3 6
```

Example Haplotypes (JSON)

File: `example_haplotypes.json`

```
{
  "Grandfather": {
    "DYS391": "10",
    "DYS389I": "13",
    "DYS439": "12",
    "DYS389II": "29",
    "DYS438": "12",
    "DYS385": "11;14"
  },
  "Father": {
    "DYS391": "10",
    "DYS389I": "13",
    "DYS439": "11",
    "DYS389II": "29",
    "DYS438": "12",
    "DYS385": "11;14"
  },
  "Son": {
    "DYS391": "10",
    "DYS389I": "13",
    "DYS439": "11",
    "DYS389II": "29",
    "DYS438": "12",
    "DYS385": "11;14"
  }
}
```

Example Config (CLI)

File: `example_config.ini`

```ini
[pedigree]
path = examples/example_pedigree.tgf
known_haplotypes = examples/example_haplotypes.json
marker_set = data/marker_sets/yfiler.csv
suspect = Son
simulation_name = Example_Analysis
user_name = Analyst Name
exclude_individuals =
two_step_mutation_factor = 0.10
stability_window = 1000
model_validity_threshold = 0.01
number_of_threads = 4
results_path = results/
bias = 0.1
```

Feedback and Support

For questions, bug reports, or suggestions:

- **GitHub**: [Repository link]
- **Email**: [Support email]
- **Documentation**: [Online docs link]

End of User Manual

MatchY - Forensic Y-STR Match Probability Analysis Empowering forensic science with rigorous statistical methods